

CREATING A LINKED LIST AS A PREAMBLE TO TREES

PROBLEM STATEMENT

You have been provided with a code. You need to write two functions, **pop_back()** – which deletes a node from tail and updates the tail- and **pop_front()** – which deletes a node from head and updates the head.

Following are the requirements:

You may watch the following video for the explanation of the code:

<https://youtu.be/vkAs048AveQ>

Assignment must be submitted in hard copy and hand written. You need to write the same code five times at least. **(2 marks each)**

```

struct Node{
    int data;
    Node *pre, *next;
    Node(){
        this->pre = nullptr;
        this->next = nullptr;
    }
};

void insert(Node *&head, Node *&tail, int value){
    Node *temp = new Node;
    temp->data = value;

    if (head == nullptr){
        head = temp;
        tail = temp;
    }
    else{
        tail->next = temp;
        temp->pre = tail;
        tail = temp;
    }
}

void showList (Node *head){
    while(head !=nullptr){
        std::cout<<head->data<<"\t";
        head = head->next;
    }
}

int main(){
    Node *head = nullptr , *tail = nullptr;

    insert (head, tail, 10);
    insert (head, tail, 20);
    insert (head, tail, 30);
    insert (head, tail, 40);
    insert (head, tail, 50);
    showList (head);
}

```